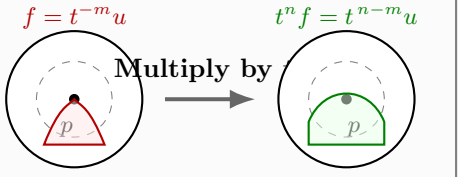
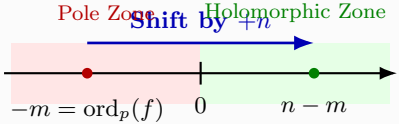
1. Local Pole-Clearing



"Multiply by the local correcting factor and ask that the singularity disappears."

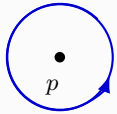
2. Order Language



$t^n f$ is holomorphic at p
$\iff \quad \mathrm{ord}_p(t^n f) \geq 0$

$$\mathrm{ord}_p(f) + \mathrm{ord}_p(t^n) \geq 0$$
$$\mathrm{ord}_{\mathbf{p}}(\mathbf{f}) + \mathbf{n} \geq \mathbf{0}$$

3. Winding-Form View

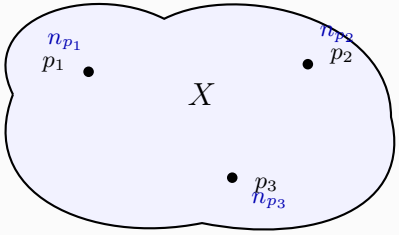


$$\mathrm{ord}_p(f) = \frac{1}{2\pi i}\int_{\gamma_p}\frac{df}{f}$$
$$n = \frac{1}{2\pi i}\int_{\gamma_p} n\,\frac{dt}{t}$$

$$\frac{1}{2\pi i}\int_{\gamma_p}\frac{d(t^n f)}{t^n f} = \mathrm{ord}_p(f) + n \geq 0$$

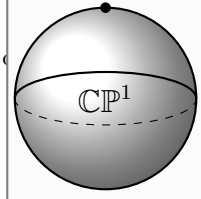
The divisor coefficient n is the local winding added to cancel the pole's negative winding.

4. Global Compression into a Divisor



Loca
der

5. Model Case on $\mathbb{CP}^1$ $(D = n[\infty])$



$f = z^m = w^{-m}$

$\times w^n$

$w^n f = w^{n-m}$ → holo at 0 → $m \leq n$

$$L(n[\infty]) = \mathrm{span}_{\mathbb{C}}\{1, z, z^2, \ldots, z^n\}$$

The allowed functions are exactly polynomials of degree $\leq n$.